GREEEN tallgrass prairie restoration: RLQ analyses

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Preparation

```
library(here)
library(tidyverse)
library(ade4)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types =
    cols(
    .default = "?",
    seeding_time = "f"
  )) %>%
  arrange(id_plot_year) %>%
```

```r
  filter(
    site == "Lux Arbor",
    richness_type == "seeded_richness",
    treatment_id %in% c("1", "2", "3", "4")
  ) %>%
  mutate(
    treatment = str_c(seeding_time, herbicide, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  filter(treatment != "unseeded_0_0") %>%
  select(
    id_plot_year, year, treatment, water_cap, cover_non_seeded,
    cover_seeded_grass
    )

traits <- read_csv(
  here("data", "processed", "data_processed_traits.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(seeded == "1", family != "Poaceae") %>%
  select(accepted_name, sla, height, seedmass) %>%
  drop_na()

species <- read_csv(
  here("data", "processed", "data_processed_species.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(plot_size == "1") %>%
  select(id_plot_year, accepted_name, abundance) %>%
  semi_join(sites, by = "id_plot_year") %>%
  semi_join(traits, by = "accepted_name") %>%
  arrange(id_plot_year)
```

Load data

Checks

```r
species %>%
  group_by(id_plot_year) %>%
  count()
```

Seeded species richness

```
## # A tibble: 81 x 2
## # Groups:   id_plot_year [81]
##    id_plot_year     n
##    <chr>        <int>
##  1 Lux_12_2015      1
##  2 Lux_12_2016      2
##  3 Lux_12_2017      3
##  4 Lux_12_2018      2
##  5 Lux_12_2019      3
##  6 Lux_12_2025      1
##  7 Lux_17_2015      1
```

```
##  8 Lux_17_2016      1
##  9 Lux_17_2017      4
## 10 Lux_17_2018      2
## # i 71 more rows
```

```
traits %>% anti_join(species, by = "accepted_name")
```

Check non-occuring seeded species

```
## # A tibble: 9 x 4
##   accepted_name                  sla height seedmass
##   <chr>                        <dbl>  <dbl>    <dbl>
## 1 Chamaecrista fasciculata     297.   0.73  0.00768
## 2 Dalea purpurea               191.   0.91  0.00231
## 3 Eryngium yuccifolium          91.7  1.01  0.00382
## 4 Helianthus occidentalis      184.   0.537 0.00221
## 5 Liatris cylindracea          110.   0.564 0.00326
## 6 Lupinus perennis             167.   0.326 0.0215
## 7 Solidago juncea               13.3  0.783 0.000412
## 8 Solidago nemoralis            12.8  0.61  0.000393
## 9 Symphyotrichum oolentangiense 14.3  0.362 0.00017
```

Finalize preparation

```
traits <- traits %>%
  semi_join(species, by = "accepted_name") %>%
  column_to_rownames(var = "accepted_name")

sites <- sites %>%
  semi_join(species, by = "id_plot_year") %>%
  column_to_rownames(var = "id_plot_year")

species <- species %>%
  pivot_wider(
    names_from = "accepted_name", values_from = "abundance", values_fill = 0
    ) %>%
  column_to_rownames(var = "id_plot_year")
```

Statistics

Descriptive analyses

```
sites %>%
  group_by(year) %>%
  count()
```

```
## # A tibble: 6 x 2
## # Groups:   year [6]
##    year     n
##   <dbl> <int>
## 1  2015     9
## 2  2016    13
## 3  2017    15
```

```
## 4  2018    13
## 5  2019    16
## 6  2025    15
```

Ordinations

```
dudi_l <- dudi.coa(species, scannf = FALSE)
dudi_r <- dudi.hillsmith(
  sites,
  row.w = dudi_l$lw,
  scannf = FALSE
  )
dudi_q <- dudi.pca(
  traits,
  row.w = dudi_l$cw,
  scannf = FALSE
  )
```

RLQ analyses

```
m1 <- rlq(
  dudiR = dudi_r,
  dudiL = dudi_l,
  dudiQ = dudi_q,
  scannf = FALSE
  )
```

Global test

```
randtest(m1, nrepet = 999, modeltype = 6)
```

```
## class: krandtest lightkrandtest
## Monte-Carlo tests
## Call: randtest.rlq(xtest = m1, nrepet = 999, modeltype = 6)
##
## Number of tests:   2
##
## Adjustment method for multiple comparisons:   none
## Permutation number:   999
##      Test       Obs   Std.Obs   Alter Pvalue
## 1 Model 2 0.8007095 2.3656510 greater  0.023
## 2 Model 4 0.8007095 0.9119105 greater  0.192
```

Traits by environment crossed table

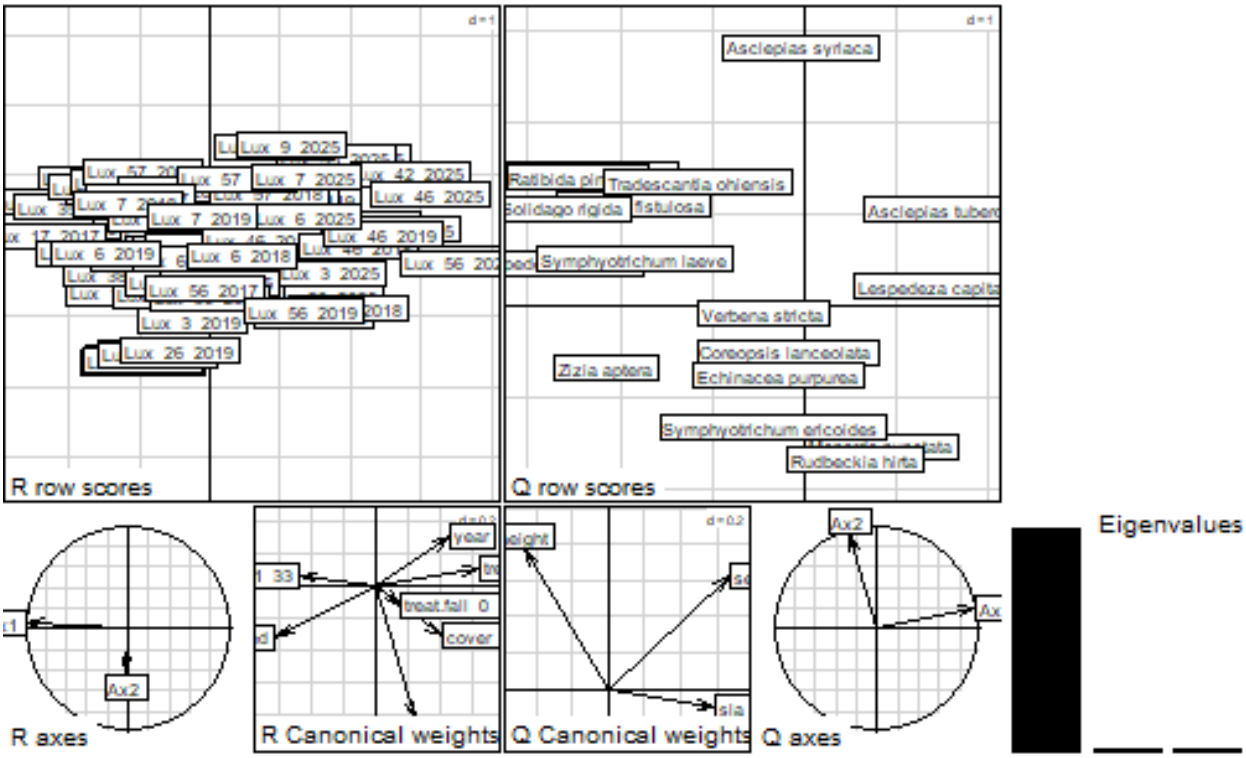
```
m1$tab
```

```
##                            sla      height   seedmass
## year                 0.20510959 -0.18107466  0.3032351
## treat.fall_0_33      0.09946235 -0.06554272  0.0805551
## treat.spring_0_33    0.32046651 -0.26259194  0.3956382
## treat.spring_1_33   -0.25536530  0.19383872 -0.2744795
## water_cap            0.12277260 -0.16360606  0.1019114
```

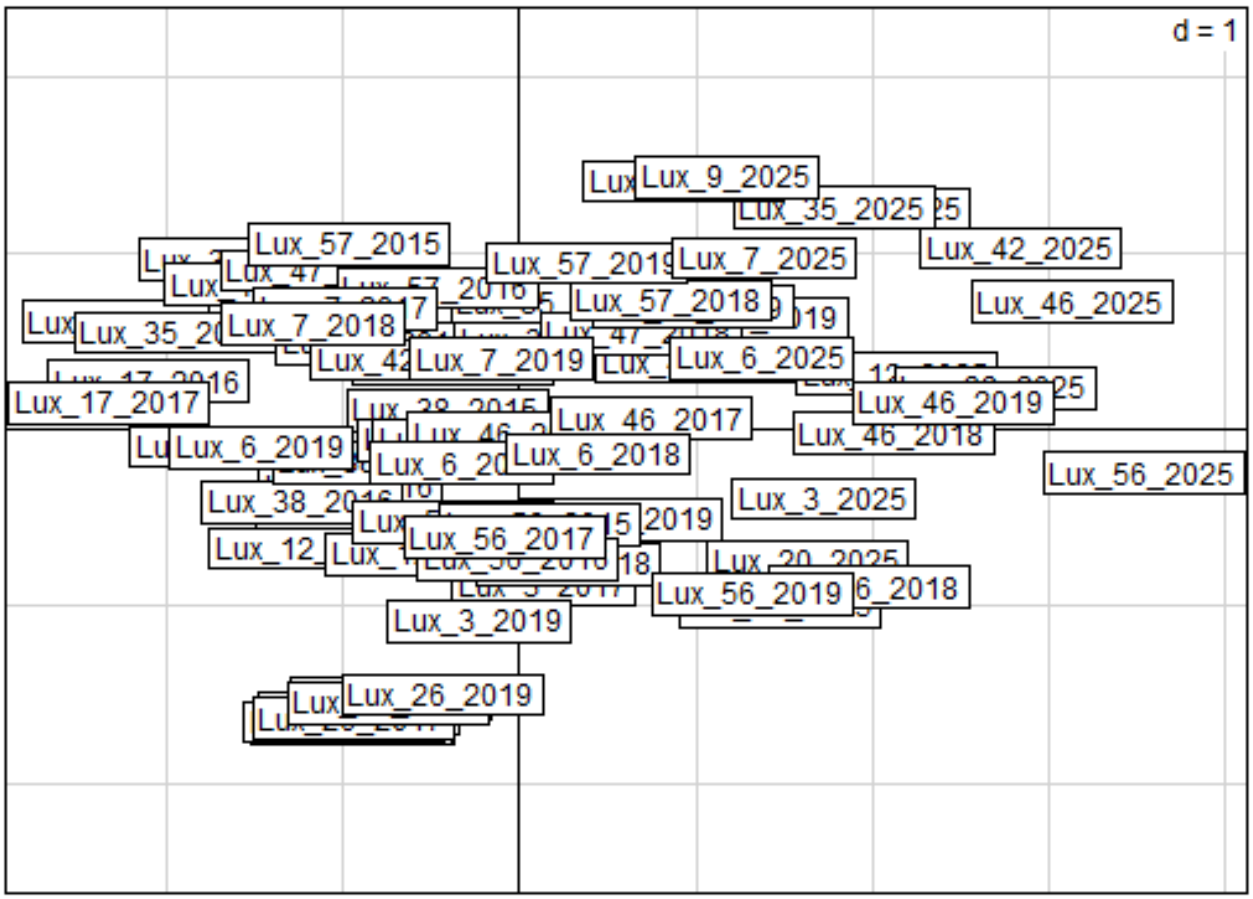
```
## cover_non_seeded   -0.33392029  0.21951993 -0.3760397
## cover_seeded_grass  0.23083585 -0.18142044  0.2183930
```

Plots

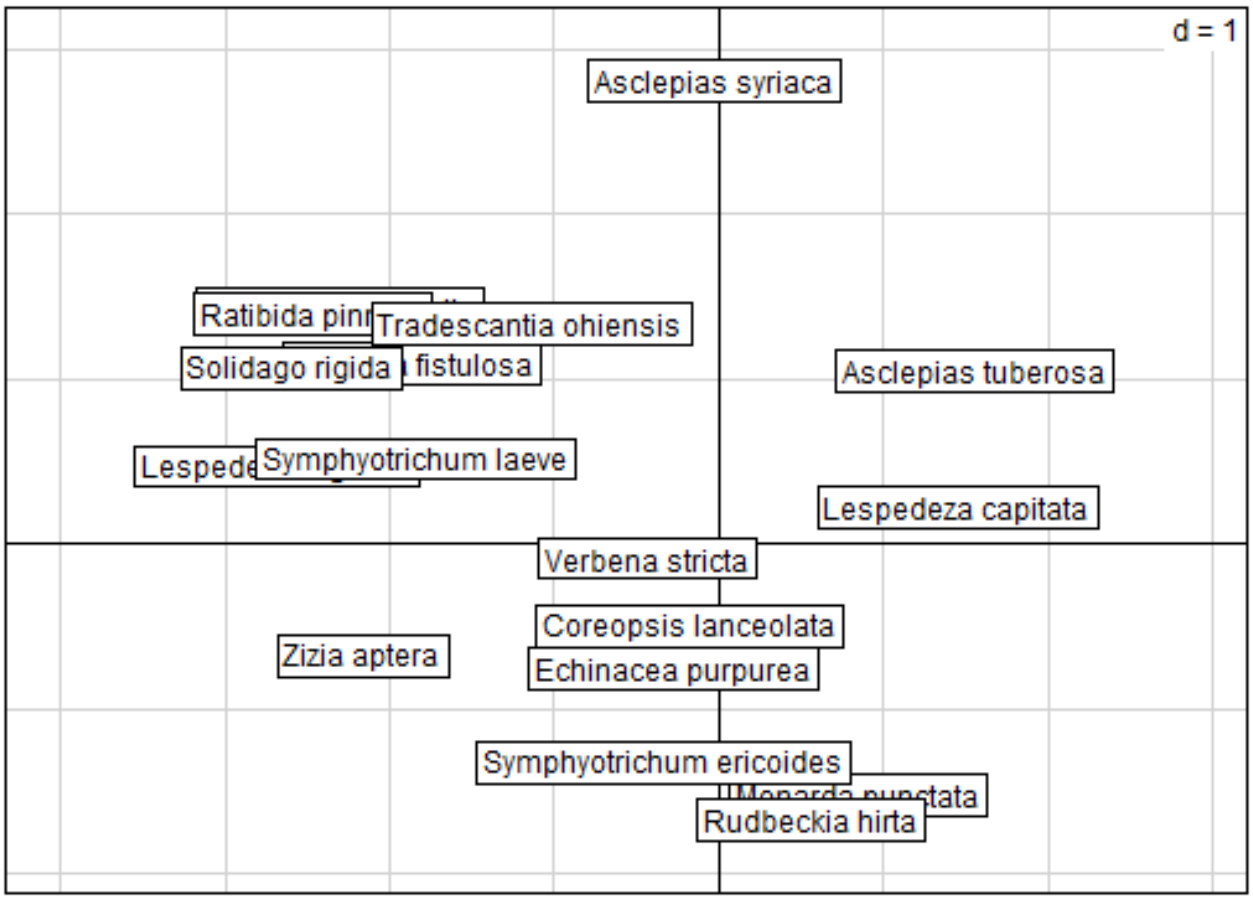
```
plot(m1)
```



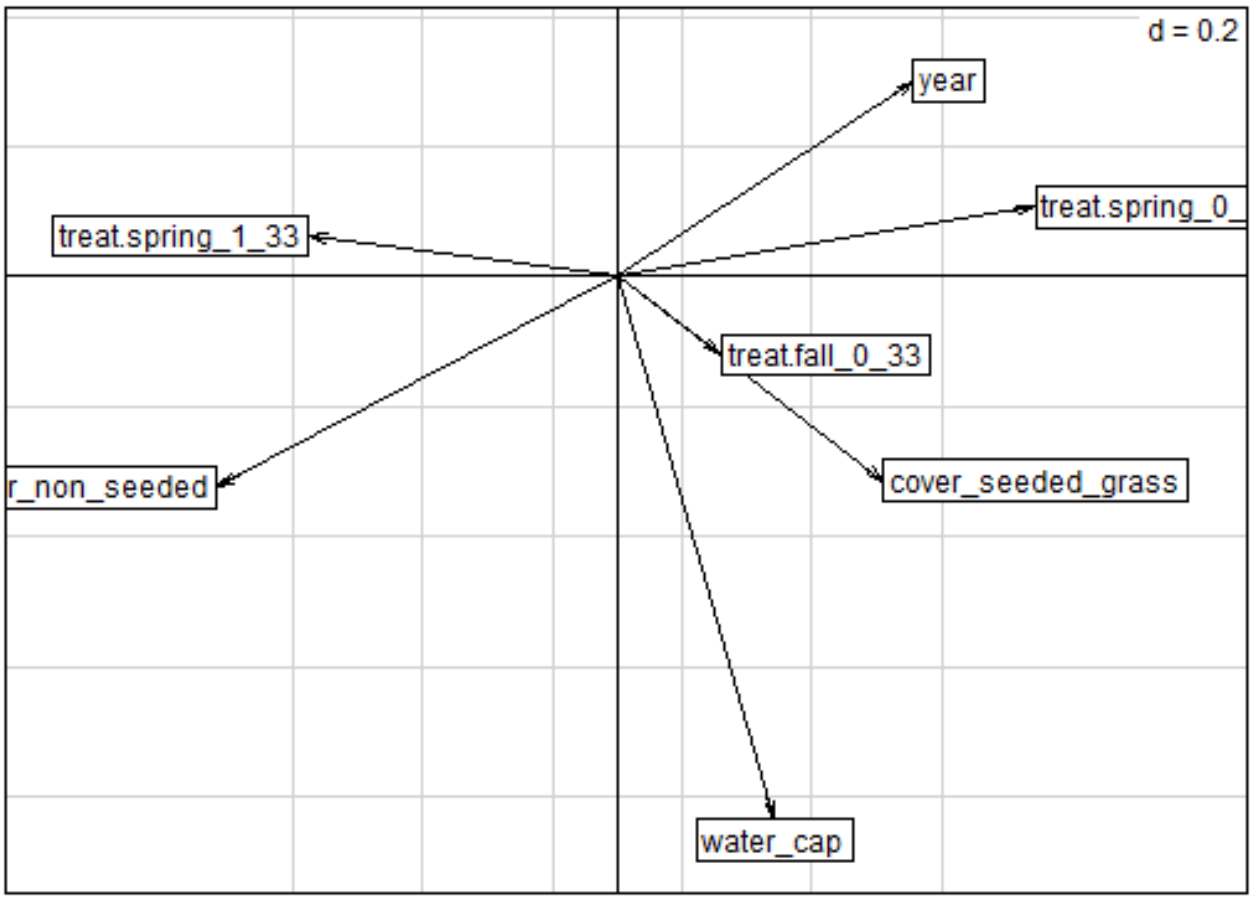

```
s.label(m1$lR)
```


d = 1
Lux_9_2025
Lux_35_2025
Lux_42_2025
Lux_57_2015
Lux_57_2019
Lux_7_2025
Lux_57_2018
Lux_46_2025
Lux_7_2018
Lux_7_2019
Lux_6_2025
Lux_17_2017
Lux_46_2019
Lux_46_2017
Lux_46_2018
Lux_6_2019
Lux_6_2018
Lux_56_2025
Lux_3_2025
Lux_56_2017
Lux_56_2019
Lux_3_2019
Lux_26_2019

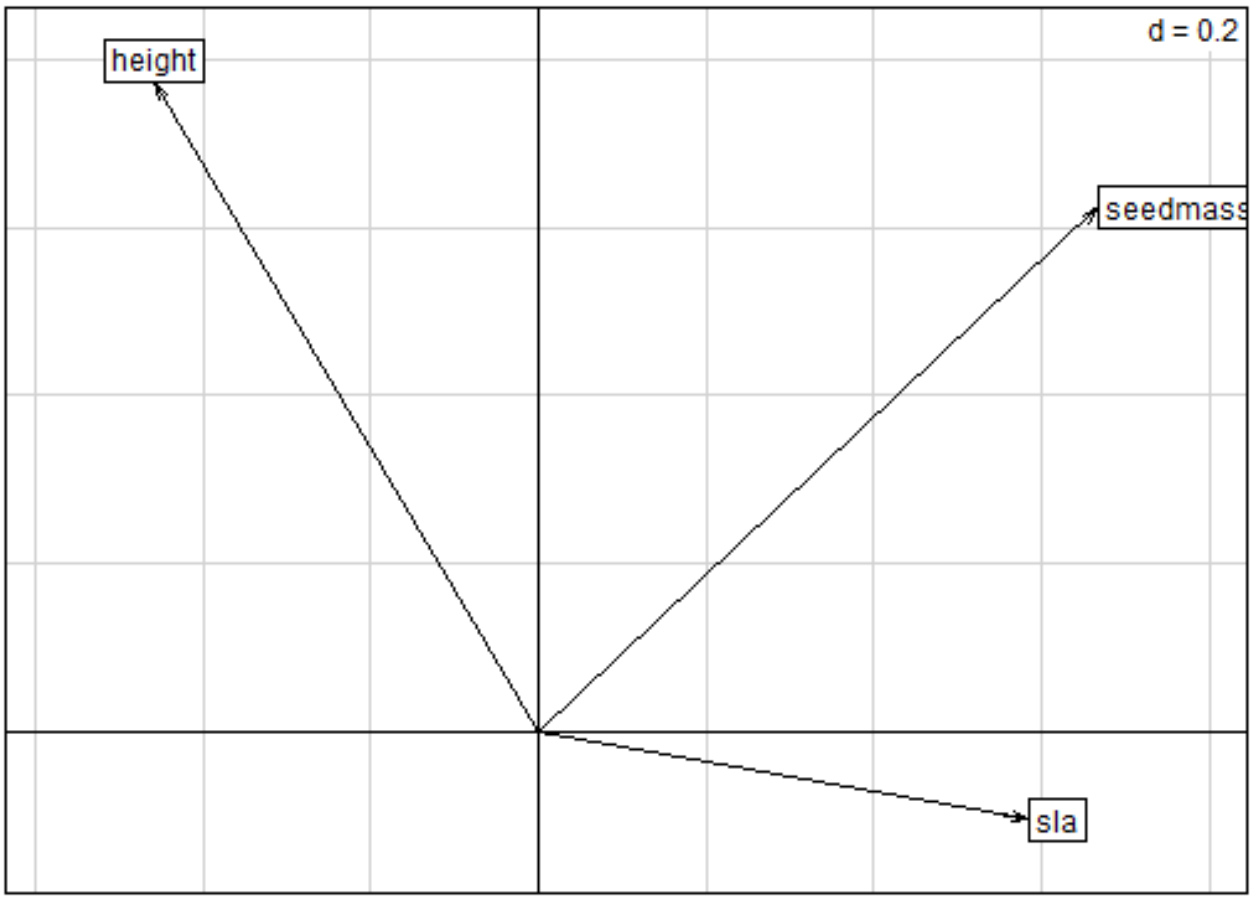
```
s.label(m1$lQ)
```


d = 1
Asclepias syriaca
Ratibida pin
Tradescantia ohiensis
Solidago rigida
fistulosa
Asclepias tuberosa
Lespede
Symphyotrichum laeve
Lespedeza capitata
Verbena stricta
Coreopsis lanceolata
Zizia aptera
Echinacea purpurea
Symphyotrichum ericoides
Monarda punctata
Rudbeckia hirta

```
s.arrow(m1$l1)
```


d = 0.2
year
treat.spring_0_
treat.spring_1_33
treat.fall_0_33
r_non_seeded
cover_seeded_grass
water_cap

```
s.arrow(m1$c1)
```



Save

Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] ade4_1.7-23     lubridate_1.9.4 forcats_1.0.0   stringr_1.5.1
```

```
##  [5] dplyr_1.1.4     purrr_1.1.0     readr_2.1.5     tidyr_1.3.1
##  [9] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0 here_1.0.1
## [13] tinytex_0.57
##
## loaded via a namespace (and not attached):
##  [1] bit_4.6.0          gtable_0.3.6       crayon_1.5.3       compiler_4.5.0
##  [5] Rcpp_1.1.0         tidyselect_1.2.1   parallel_4.5.0     scales_1.4.0
##  [9] yaml_2.3.10        fastmap_1.2.0      R6_2.6.1           generics_0.1.4
## [13] knitr_1.50         MASS_7.3-65        rprojroot_2.0.4    pillar_1.11.0
## [17] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.6        utf8_1.2.6
## [21] stringi_1.8.7      xfun_0.52          bit64_4.6.0-1      timechange_0.3.0
## [25] cli_3.6.5          withr_3.0.2        magrittr_2.0.3     digest_0.6.37
## [29] grid_4.5.0         vroom_1.6.5        rstudioapi_0.17.1  hms_1.1.3
## [33] lifecycle_1.0.4    vctrs_0.6.5        evaluate_1.0.4     glue_1.8.0
## [37] farver_2.1.2       rmarkdown_2.29     tools_4.5.0        pkgconfig_2.0.3
## [41] htmltools_0.5.8.1
```